Physics for Scientists and Engineers Storyline

Tenth Edition

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Chapter 23

Continuous Charge Distributions and Gauss's Law

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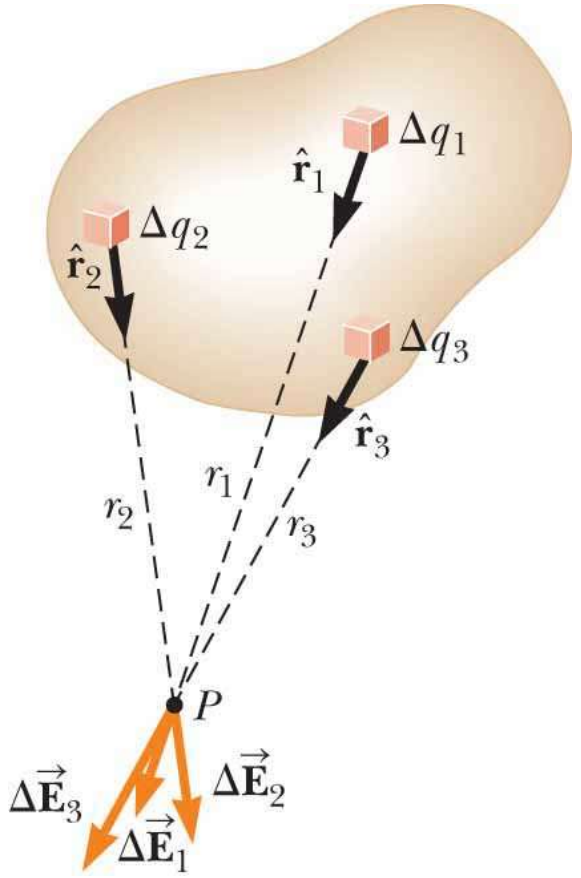
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Electric Field of a Continuous Charge Distribution

$$\Delta\vec{\mathbf{E}} = k_e \frac{\Delta q}{r^2}\hat{\mathbf{r}}$$

$$\vec{\mathbf{E}} \approx k_e \sum_i \frac{\Delta q_i}{r_i^2}\hat{\mathbf{r}}_i$$

$$\vec{\mathbf{E}} = k_e \lim_{\Delta q_i \to 0} \sum_i \frac{\Delta q_i}{r_i^2}\hat{\mathbf{r}}_i = k_e \int \frac{dq}{r^2}\hat{\mathbf{r}}$$



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Charge Density

Charge density:

$$\rho \equiv \frac{Q}{V} \qquad dq = \rho\, dV$$

$$\sigma \equiv \frac{Q}{A} \qquad dq = \sigma\, dA$$

$$\lambda \equiv \frac{Q}{\ell} \qquad dq = \lambda\, d\ell$$

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Problem-Solving Strategy: Calculating the Electric Field

1. **Conceptualize**
2. **Categorize**
3. **Analyze**
4. **Finalize**

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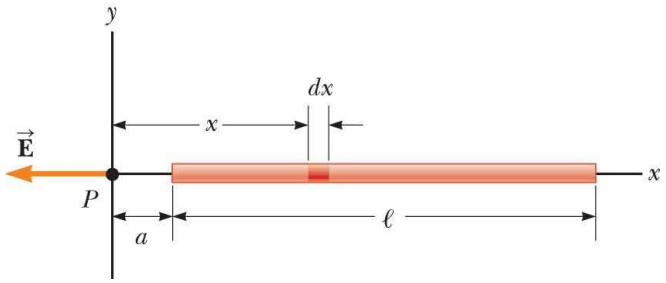
Example 23.1: Electric Field Due to a Charged Rod (1 of 2)

A rod of length ℓ has a uniform positive charge per unit length λ and a total charge Q. Calculate the electric field at a point P that is located along the long axis of the rod and a distance a from one end.

$$dE = k_e \frac{dq}{x^2} = k_e \frac{\lambda dx}{x^2}$$

$$E = \int_a^{\ell+a} k_e \lambda \frac{dx}{x^2}$$

$$E = k_e \lambda \int_a^{\ell+a} \frac{dx}{x^2} = k_e \lambda \left[-\frac{1}{x} \right]_a^{\ell+a}$$



$$E = k_e \frac{Q}{\ell} \left(\frac{1}{a} - \frac{1}{\ell + a} \right) = \boxed{\frac{k_e Q}{a(\ell + a)}}$$

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Example 23.1: Electric Field Due to a Charged Rod (2 of 2)

Suppose point *P* is very far away from the rod. What is the nature of the electric field at such a point?

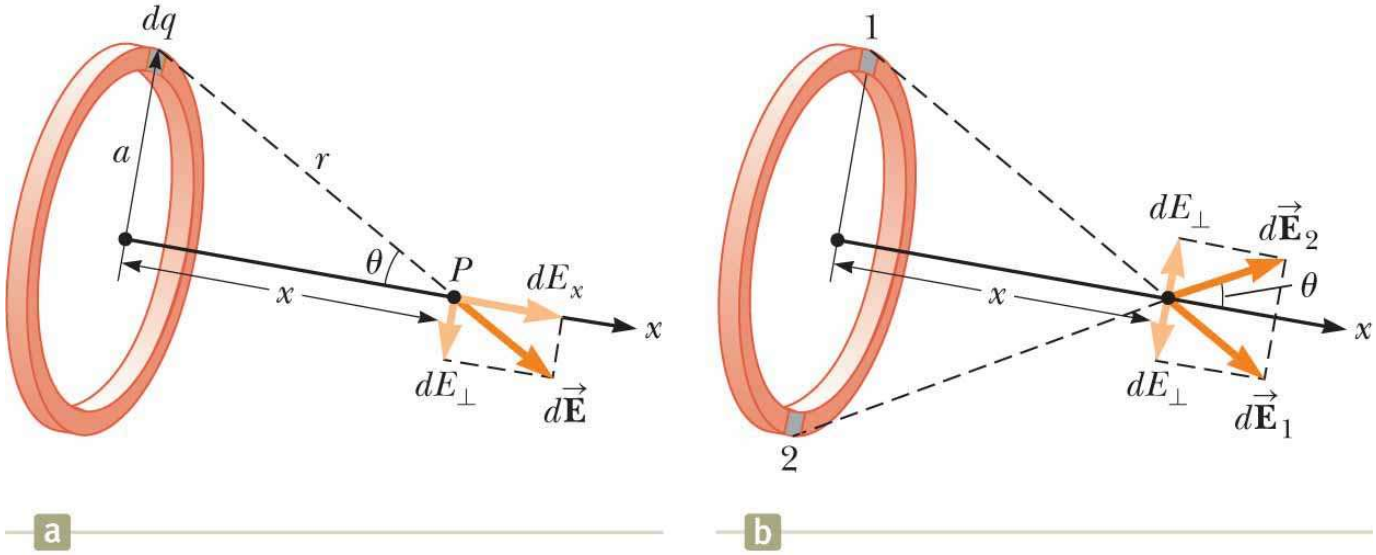
$$E = \frac{k_e Q}{a(\ell + a)} \approx \frac{k_e Q}{a^2}$$

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Example 23.2: Electric Field Due to a Uniform Ring of Charge (1 of 3)

A ring of radius a carries a uniformly distributed positive total charge *Q*. Calculate the electric field due to the ring at a point *P* lying a distance *x* from its center along the central axis perpendicular to the plane of the ring



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Example 23.2: Electric Field Due to a Uniform Ring of Charge (2 of 3)

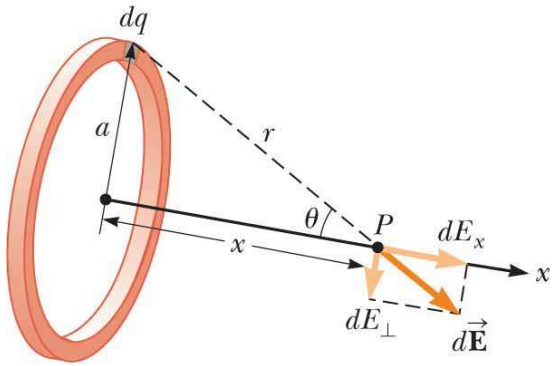
$$dE_x = k_e \frac{dq}{r^2}\cos\theta = k_e \frac{dq}{a^2 + x^2}\cos\theta$$

$$\cos\theta = \frac{x}{r} = \frac{x}{\left(a^2 + x^2\right)^{1/2}}$$

$$dE_x = k_e \frac{dq}{a^2 + x^2}\left[\frac{x}{\left(a^2 + x^2\right)^{1/2}}\right] = \frac{k_e x}{\left(a^2 + x^2\right)^{3/2}}dq$$

$$E_x = \int \frac{k_e x}{\left(a^2 + x^2\right)^{3/2}}dq = \frac{k_e x}{\left(a^2 + x^2\right)^{3/2}}\int dq$$

$$E = \boxed{\frac{k_e x}{\left(a^2 + x^2\right)^{3/2}}Q}$$



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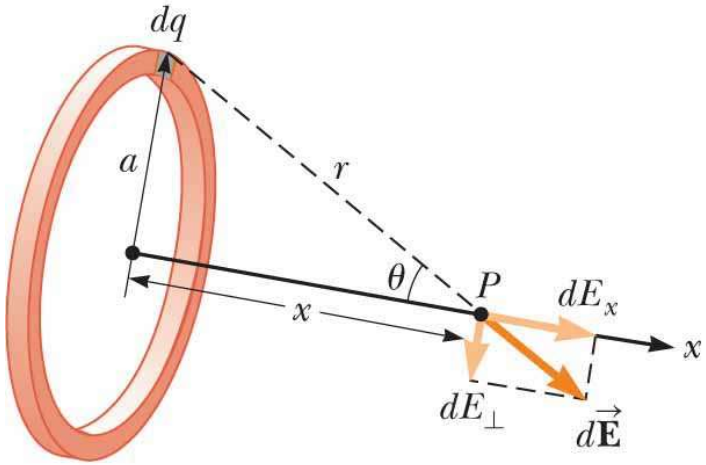
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Example 23.2: Electric Field Due to a Uniform Ring of Charge (3 of 3)

Suppose a negative charge is placed at the center of the ring in the figure and displaced slightly by a distance $x << a$ along the x axis. When the charge is released, what type of motion does it exhibit?

$$E_x = \frac{k_e Q}{a^3} x$$

$$F_x = -\frac{k_e Q q}{a^3} x$$



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Moongate and Electric Fields

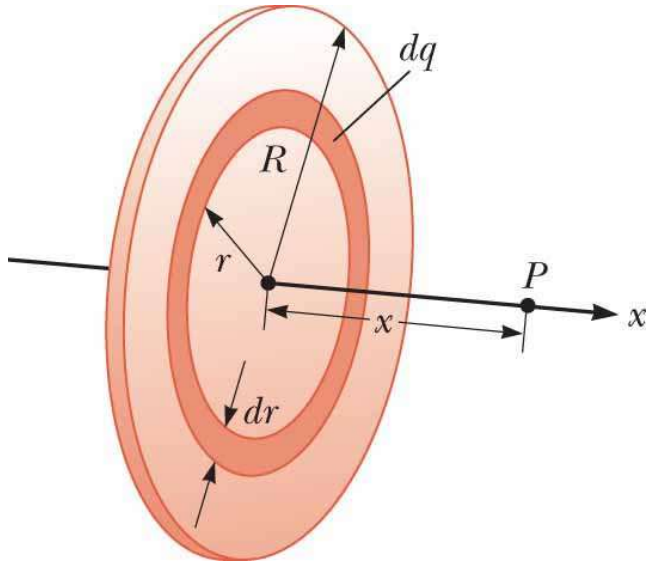


$$E = \frac{k_e x}{\left(a^2 + x^2\right)^{3/2}} Q$$

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Example 23.3: The Electric Field of a Uniformly Charged Disk (1 of 4)

A disk of radius R has a uniform surface charge density σ. Calculate the electric field at a point P that lies along the central perpendicular axis of the disk and a distance x from the center of the disk



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Example 23.3: The Electric Field of a Uniformly Charged Disk (2 of 4)

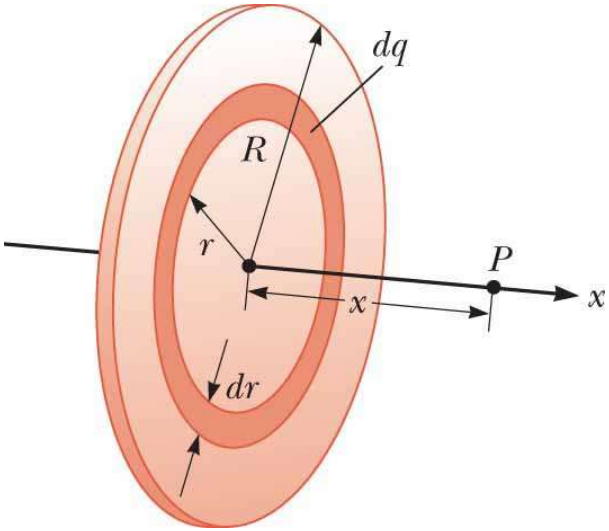
$$dq = \sigma\, dA = \sigma\left(2\pi r\, dr\right) = 2\pi\sigma r\, dr$$

$$dE_x = \frac{k_e x}{\left(a^2 + x^2\right)^{3/2}} dq \Rightarrow \frac{k_e x}{\left(r^2 + x^2\right)^{3/2}}\left(2\pi\sigma r\, dr\right)$$

$$E_x = k_e x\pi\sigma \int_0^R \frac{2r\, dr}{\left(r^2 + x^2\right)^{3/2}}$$

$$= k_e x\pi\sigma \int_0^R \left(r^2 + x^2\right)^{-3/2} d\left(r^2\right)$$

$$= k_e x\pi\sigma \left[\frac{\left(r^2 + x^2\right)^{-1/2}}{-1/2}\right]_0^R = \boxed{2\pi k_e \sigma\left[1 - \frac{x}{\left(R^2 + x^2\right)^{1/2}}\right]}$$



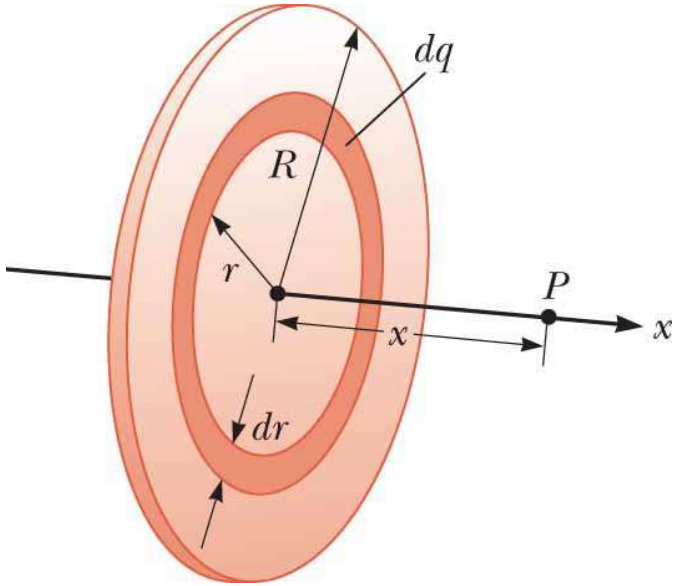
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Example 23.3: The Electric Field of a Uniformly Charged Disk (3 of 4)

$$E_x = 2\pi k_e \sigma \left[1 - \frac{x}{\left(R^2 + x^2 \right)^{1/2}} \right]$$

$$\text{for } x << R: E = 2\pi k_e \sigma = \frac{\sigma}{2\varepsilon_0}$$



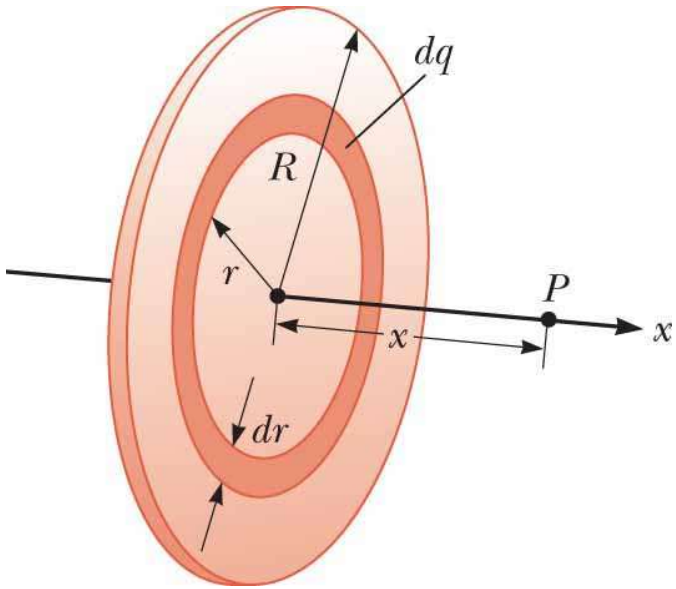
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Example 23.3: The Electric Field of a Uniformly Charged Disk (4 of 4)

What if we let the radius of the disk grow so that the disk becomes an infinite plane of charge?

$$E = 2\pi k_e \sigma = \frac{\sigma}{2\varepsilon_0}$$

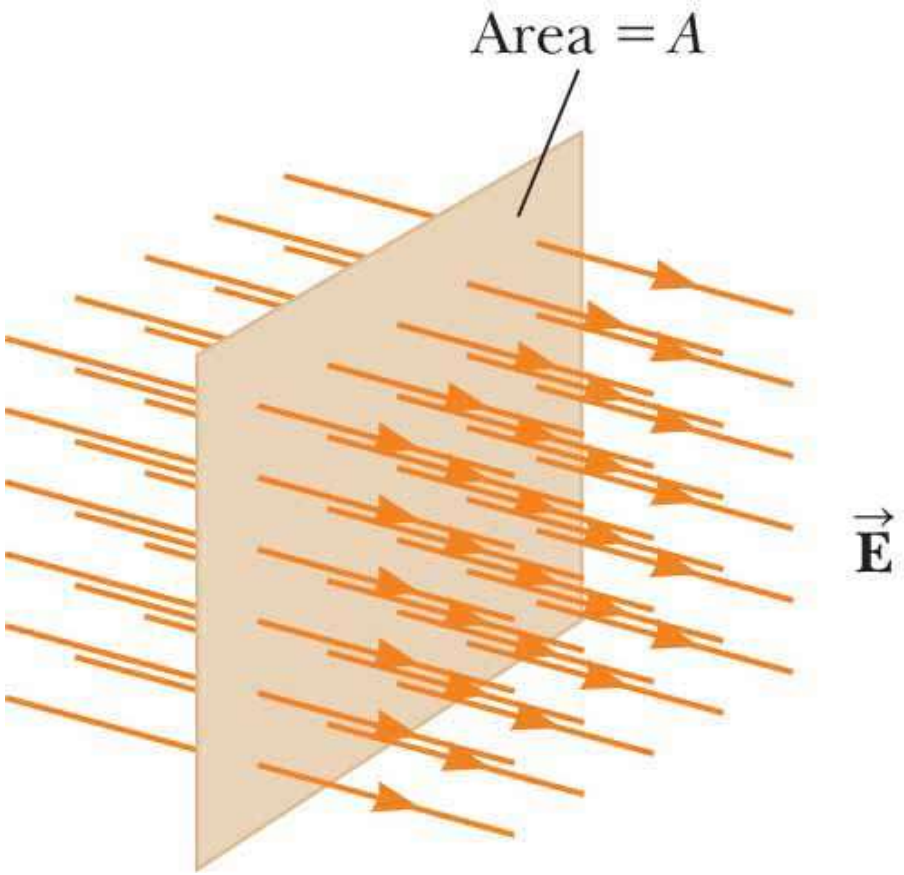


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Electric Flux (1 of 4)

$$\Phi_E = EA$$



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Electric Flux (2 of 4)

$$A = \ell w$$

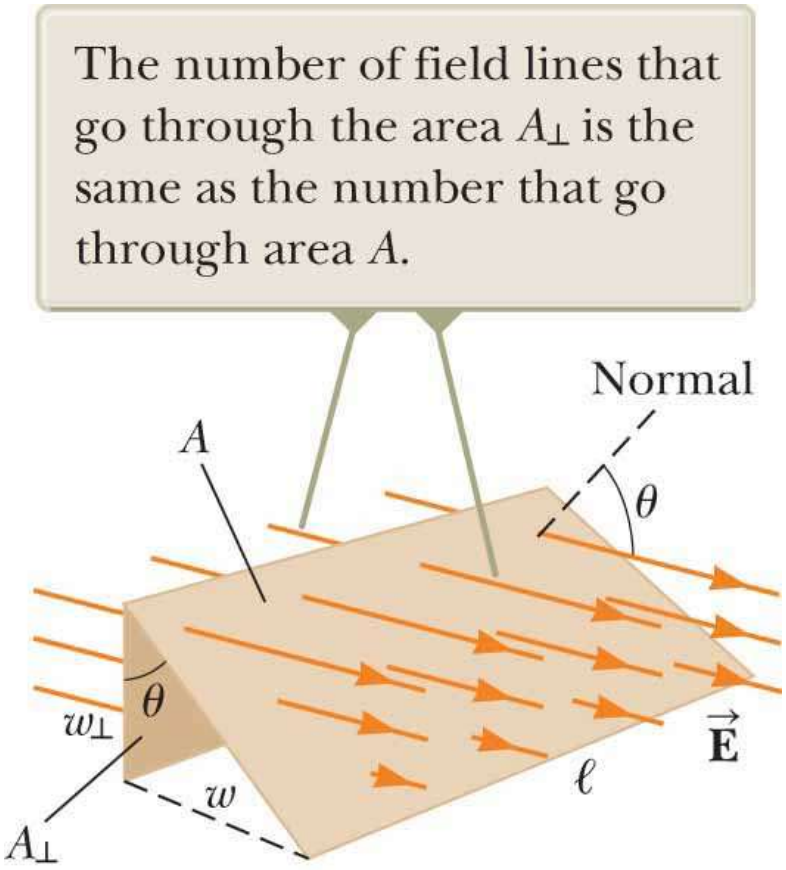
$$w_\perp = w\cos\theta$$

$$A_\perp = \ell w_\perp = \ell w\cos\theta$$

$$A_\perp = A\cos\theta$$

$$\Phi_E = EA_\perp = EA\cos\theta$$

$$\Phi_E = \left(E\cos\theta\right)A = E_n A$$



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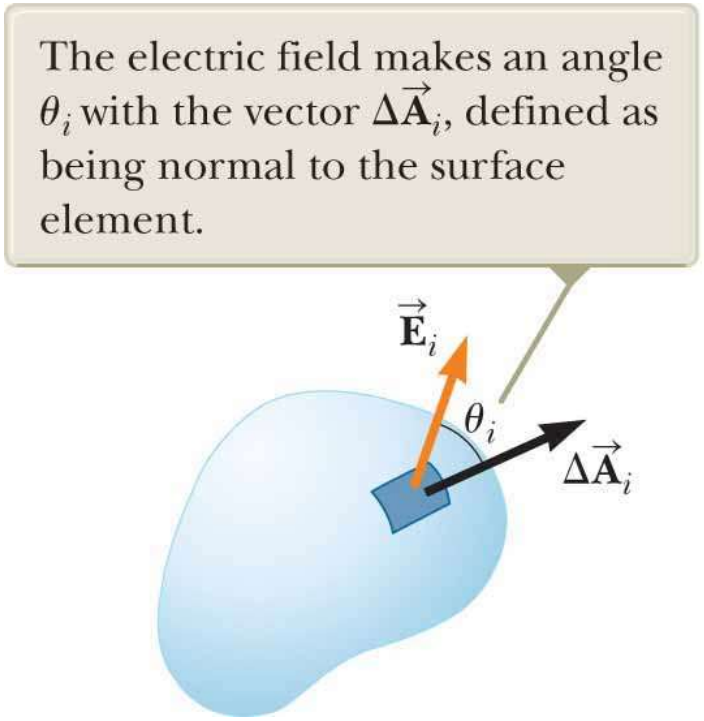
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Electric Flux (3 of 4)

$$\Phi_{E,i} = E_i \Delta A_i \cos\theta_i = \vec{\mathbf{E}}_i \cdot \Delta\vec{\mathbf{A}}_i$$

$$\Phi_E \approx \vec{\mathbf{E}}_i \cdot \Delta\vec{\mathbf{A}}_i$$

$$\Phi_E \equiv \int_{\text{surface}} \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}}$$

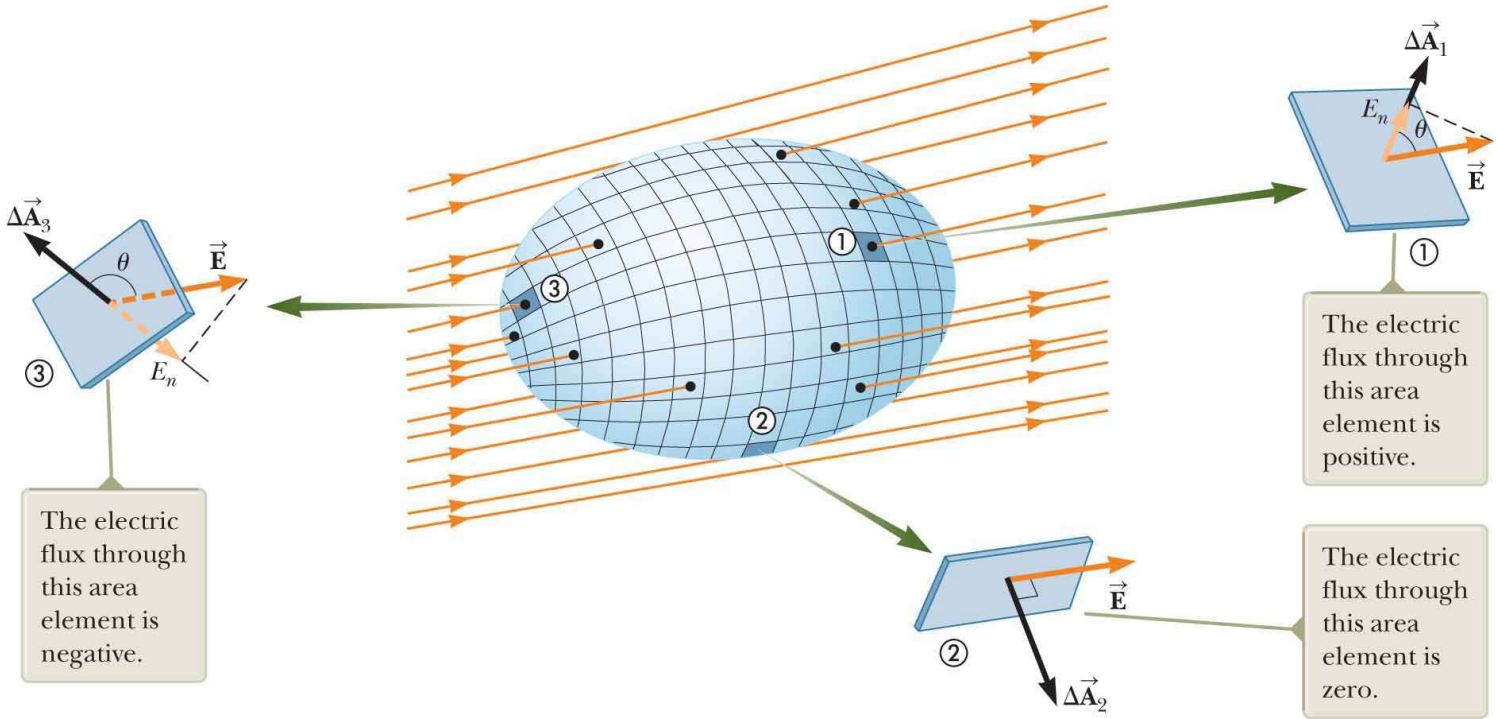


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Electric Flux (4 of 4)

$$\Phi_E = \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \oint E_n \, dA$$



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Quick Quiz 23.1 (1 of 2)

Suppose a point charge is located at the center of a spherical surface. The electric field at the surface of the sphere and the total flux through the sphere are determined. Now the radius of the sphere is halved. What happens to the flux through the sphere and the magnitude of the electric field at the surface of the sphere?

(a) The flux and field both increase.

(b) The flux and field both decrease.

(c) The flux increases, and the field decreases.

(d) The flux decreases, and the field increases.

(e) The flux remains the same, and the field increases.

(f) The flux decreases, and the field remains the same.

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Quick Quiz 23.1 (2 of 2)

Suppose a point charge is located at the center of a spherical surface. The electric field at the surface of the sphere and the total flux through the sphere are determined. Now the radius of the sphere is halved. What happens to the flux through the sphere and the magnitude of the electric field at the surface of the sphere?

(a) The flux and field both increase.

(b) The flux and field both decrease.

(c) The flux increases, and the field decreases.

(d) The flux decreases, and the field increases.

(e) The flux remains the same, and the field increases.

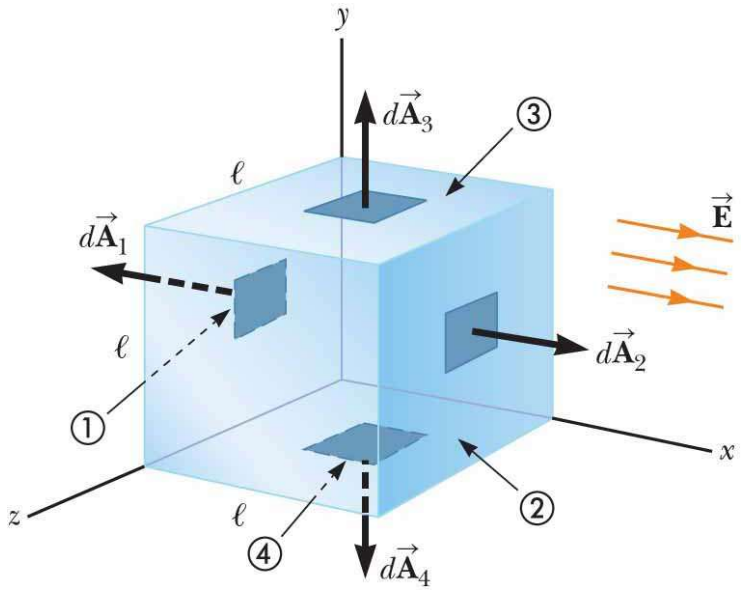
(f) The flux decreases, and the field remains the same.

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Example 23.4: Flux Through a Cube (1 of 2)

Consider a uniform electric field $\vec{\mathbf{E}}$ oriented in the *x* direction in empty space. A cube of edge length ℓ is placed in the field, oriented as shown in the figure. Find the net electric flux through the surface of the cube.



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Example 23.4: Flux Through a Cube (2 of 2)

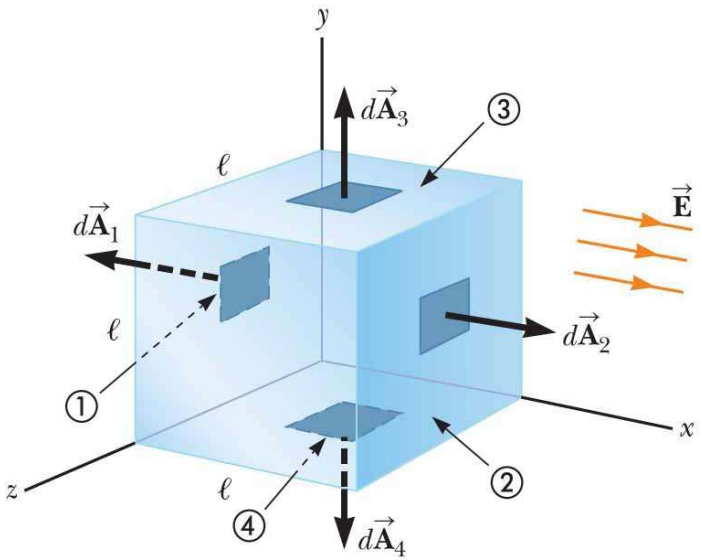
$$\Phi_E = \int_1 \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} + \int_2 \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}}$$

$$\int_1 \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \int_1 E\left(\cos 180°\right) dA = -\int_1 E\, dA = -EA = -E\ell^2$$

$$\int_2 \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \int_1 E\left(\cos 0°\right) dA$$

$$= E\int_2 dA = +EA = E\ell^2$$

$$\Phi_E = -E\ell^2 + E\ell^2 + 0 + 0 + 0 + 0 = \boxed{0}$$



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Gauss's Law

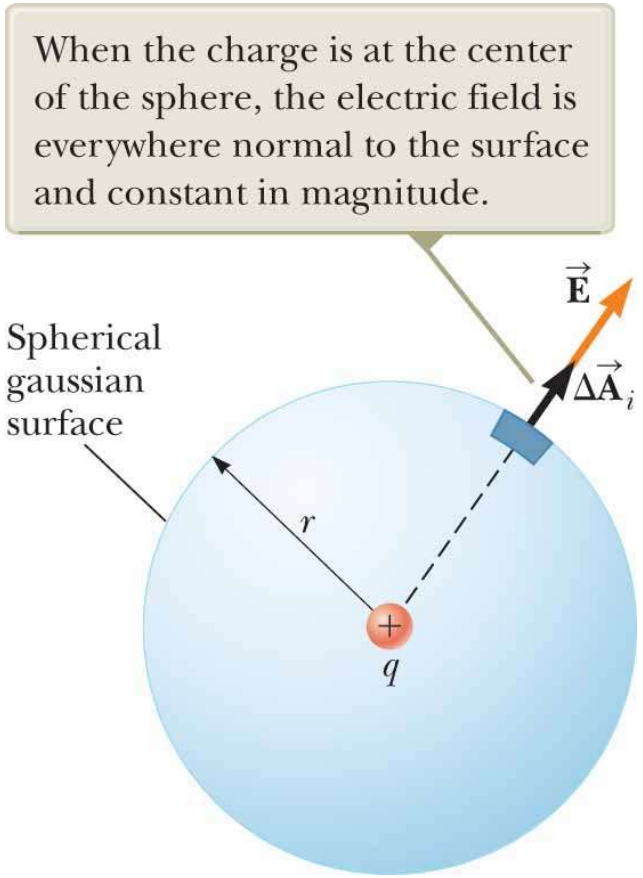
$$E = \frac{k_e q}{r^2}$$

$$\vec{\mathbf{E}} \cdot \Delta\vec{\mathbf{A}}_i = E\Delta A_i$$

$$\Phi_E = \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \oint E dA = E \oint dA$$

$$\Phi_E = \frac{k_e q}{r^2}\left(4\pi r^2\right) = 4\pi k_e q$$

$$\Phi_E = \frac{q}{\varepsilon_0}$$

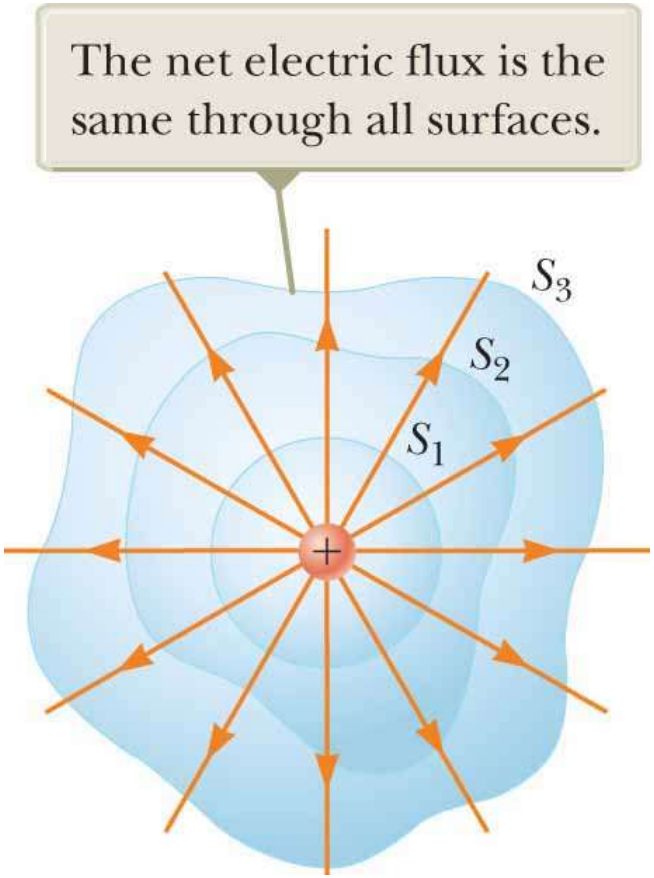


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Gaussian Surface and Flux (1 of 3)

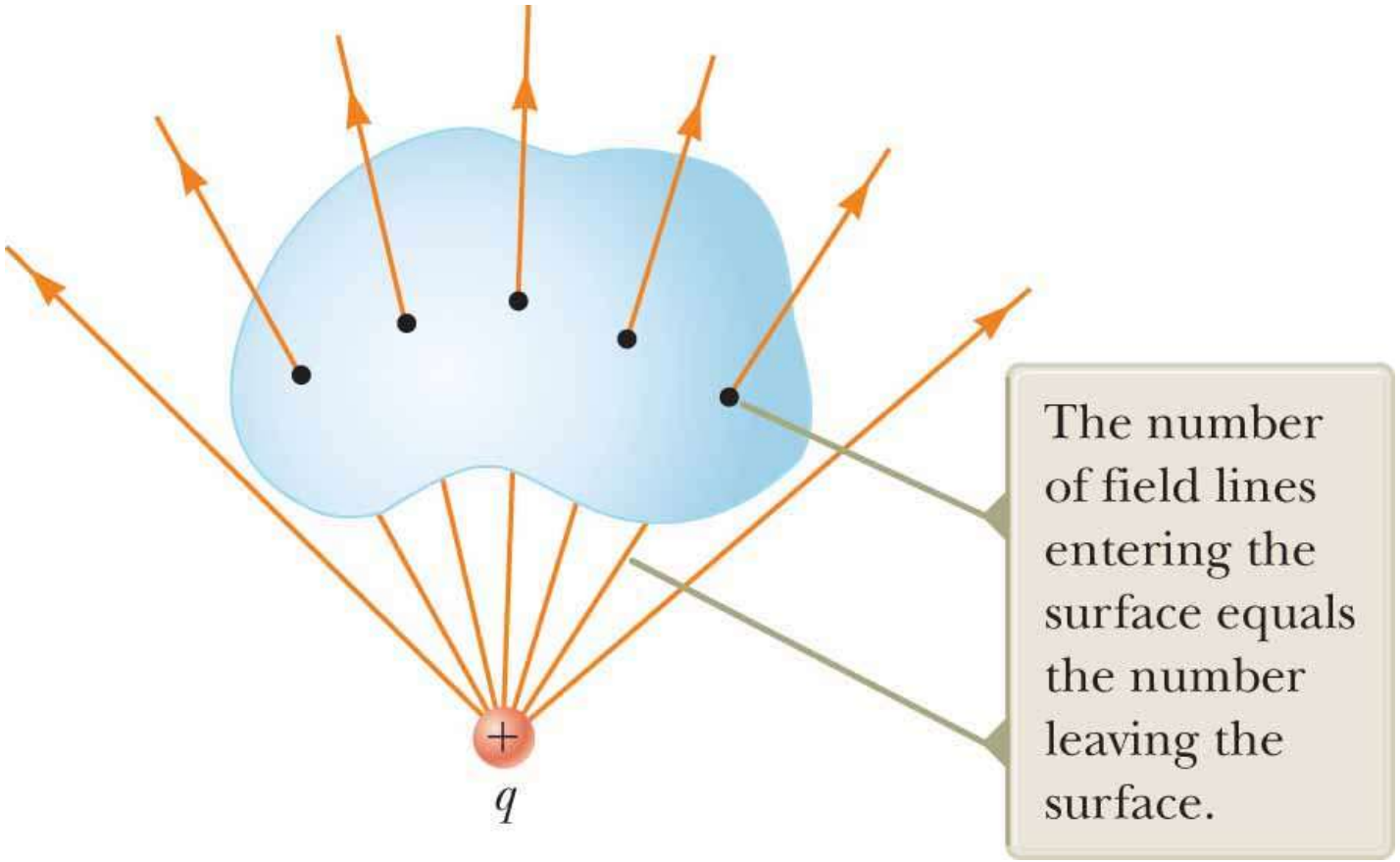
Net flux through any closed surface surrounding a point charge q is given by q/ε_0 and is independent of the shape of that surface.



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Gaussian Surface and Flux (2 of 3)



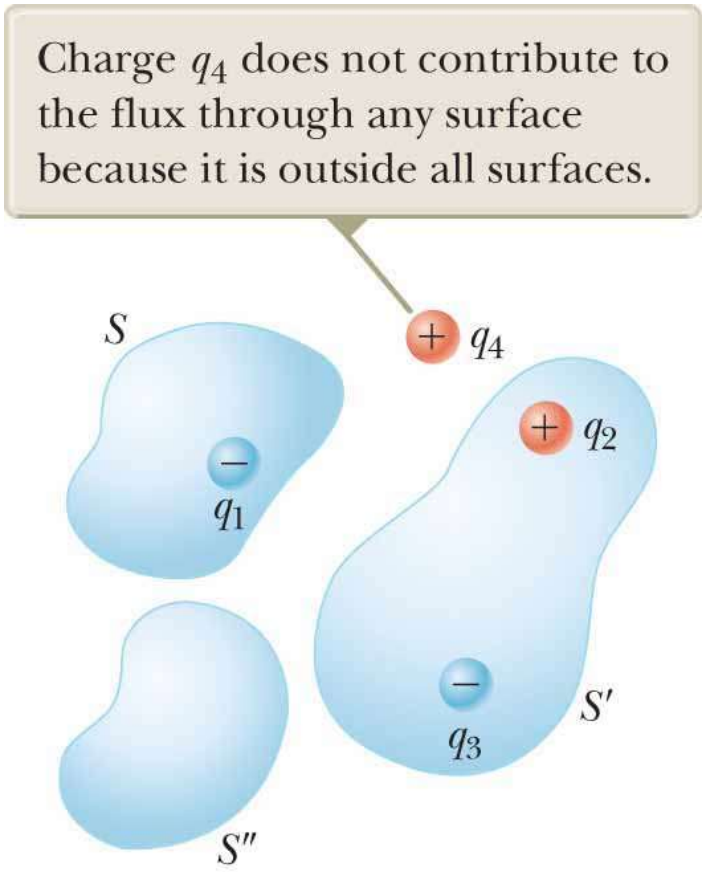
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Gaussian Surface and Flux (3 of 3)

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \oint \left(\vec{\mathbf{E}}_1 + \vec{\mathbf{E}}_2 + \ldots \right) \cdot d\vec{\mathbf{A}}$$

$$\Phi_E = \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{\text{in}}}{\varepsilon_0}$$



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Quick Quiz 23.2 (1 of 2)

If the net flux through a gaussian surface is zero, the following four statements could be true. Which of the statements must be true?

(a) There are no charges inside the surface.

(b) The net charge inside the surface is zero.

(c) The electric field is zero everywhere on the surface.

(d) The number of electric field lines entering the surface equals the number leaving the surface.

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Quick Quiz 23.2 (2 of 2)

If the net flux through a gaussian surface is zero, the following four statements could be true. Which of the statements must be true?

(a) There are no charges inside the surface.

(b) The net charge inside the surface is zero.

(c) The electric field is zero everywhere on the surface.

(d) The number of electric field lines entering the surface equals the number leaving the surface.

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Conceptual Example 23.5: Flux Due to a Point Charge

A spherical gaussian surface surrounds a point charge q. Describe what happens to the total flux through the surface if

(A) the charge is tripled,

(B) the radius of the sphere is doubled,

(C) the surface is changed to a cube, and

(D) the charge is moved to another location inside the surface.

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Application of Gauss's Law to Various Charge Distributions

$$\Phi_E = \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{\text{in}}}{\varepsilon_0}$$

1. $\vec{\mathbf{E}}$ constant over the portion of surface.
2. $\vec{\mathbf{E}}$ and $d\vec{\mathbf{A}}$ parallel: dot product = $E\,dA$
3. $\vec{\mathbf{E}}$ and $d\,\vec{\mathbf{A}}$ perpendicular: dot product = 0
4. $\vec{\mathbf{E}}$ = 0 over portion of surface

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Example 23.6: A Spherically Symmetric Charge Distribution (1 of 3)

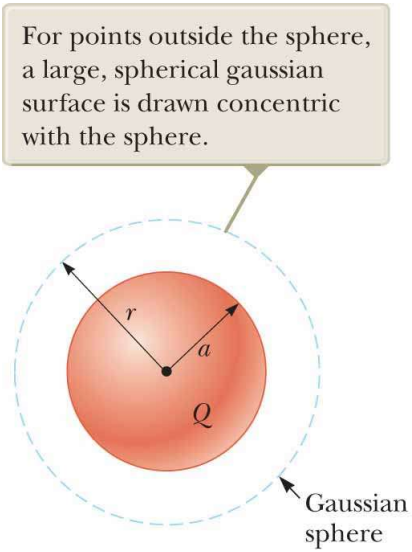
An insulating solid sphere of radius a has a uniform volume charge density ρ and carries a total positive charge Q.

(A) Calculate the magnitude of the electric field at a point outside the sphere.

$$\Phi_E = \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \oint E\, dA = \frac{Q}{\varepsilon_0}$$

$$\oint E\, dA = E \oint dA = E\left(4\pi r^2\right) = \frac{Q}{\varepsilon_0}$$

$$E = \frac{Q}{4\pi r^2 \varepsilon_0} = k_e \frac{Q}{r^2} \quad \left(\text{for } r > a\right)$$



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Example 23.6: A Spherically Symmetric Charge Distribution (2 of 3)

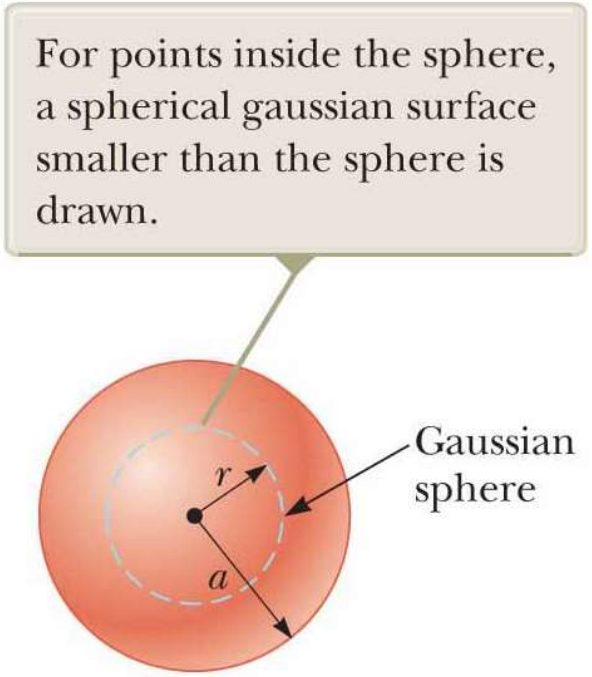
(B) Find the magnitude of the electric field at a point inside the sphere.

$$q_{\text{in}} = \rho V' = \rho\left(\frac{4}{3}\pi r^3\right)$$

$$\oint E\,dA = E\oint dA = E\left(4\pi r^2\right) = \frac{q_{\text{in}}}{\varepsilon_0}$$

$$E = \frac{q_{\text{in}}}{4\pi r^2\varepsilon_0} = \frac{\rho\left(\frac{4}{3}\pi r^3\right)}{4\pi\varepsilon_0 r^2} = \frac{\rho}{3\varepsilon_0}r$$

$$E = \frac{\frac{Q}{\frac{4}{3}\pi a^3}}{3\left(\frac{1}{4\pi k_e}\right)}r = k_e\frac{Q}{a^3}r \quad (\text{for } r < a)$$



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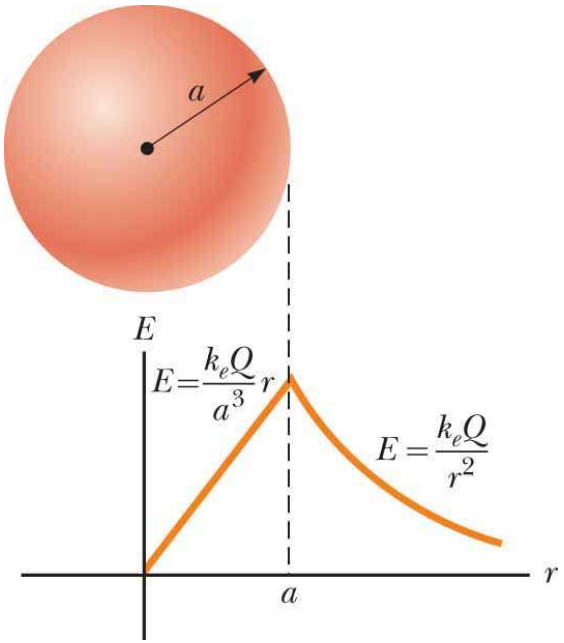
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Example 23.6: A Spherically Symmetric Charge Distribution (3 of 3)

Suppose the radial position $r = a$ is approached from inside the sphere and from outside. Do we obtain the same value of the electric field from both directions?

$$E = \lim_{r \to a} \left(k_e \frac{Q}{r^2} \right) = k_e \frac{Q}{a^2}$$

$$E = \lim_{r \to a} \left(k_e \frac{Q}{a^3} r \right) = k_e \frac{Q}{a^3} a = k_e \frac{Q}{a^2}$$

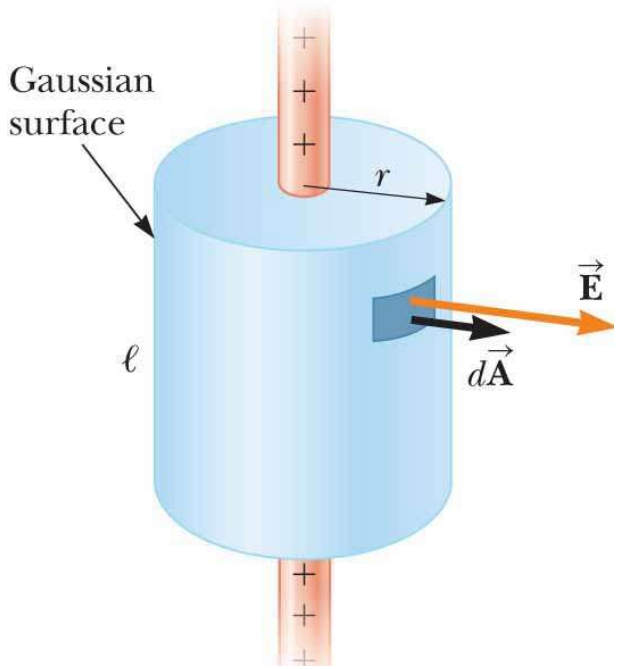


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Example 23.7: A Cylindrically Symmetric Charge Distribution (1 of 3)

Find the electric field a distance *r* from a line of positive charge of infinite length and constant charge per unit length λ.



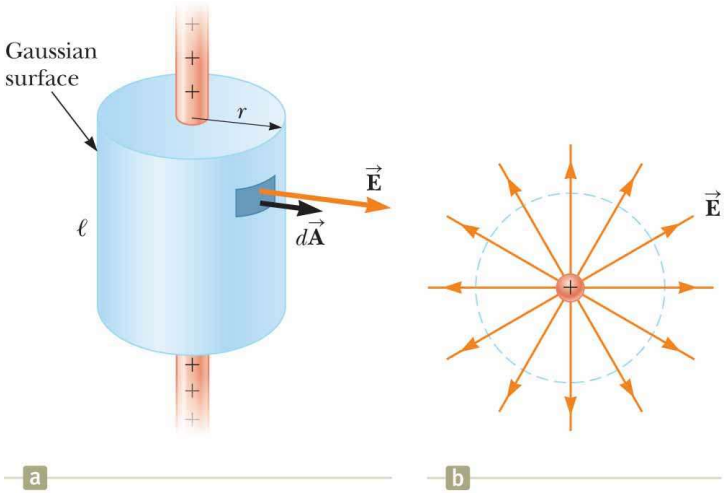
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Example 23.7: A Cylindrically Symmetric Charge Distribution (2 of 3)

$$\Phi_E = \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = E \oint dA = EA = \frac{q_{\text{in}}}{\varepsilon_0} = \frac{\lambda \ell}{\varepsilon_0}$$

$$E(2\pi r \ell) = \frac{\lambda \ell}{\varepsilon_0}$$



$$E = \frac{\lambda}{2\pi\varepsilon_0 r} = \boxed{2k_e \frac{\lambda}{r}}$$

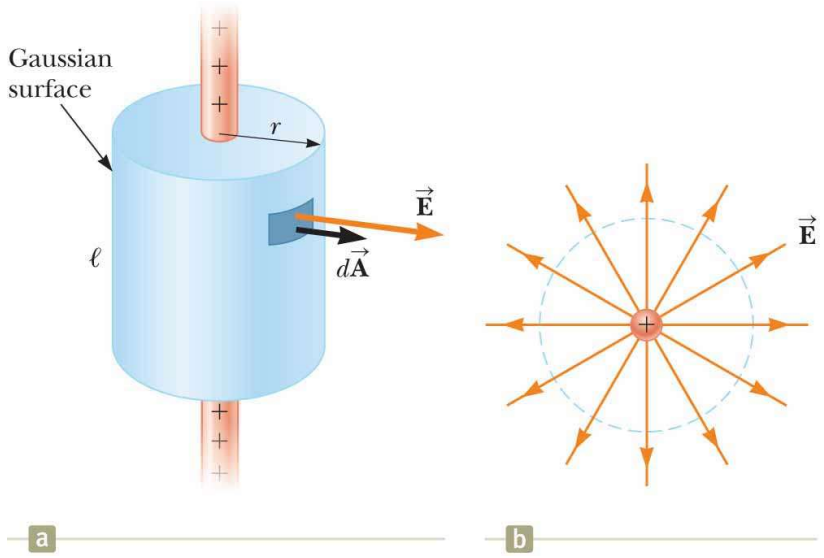
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Example 23.7: A Cylindrically Symmetric Charge Distribution (3 of 3)

What if the line segment in this example were not infinitely long?

$$E \neq 2k_e \frac{\lambda}{r}$$



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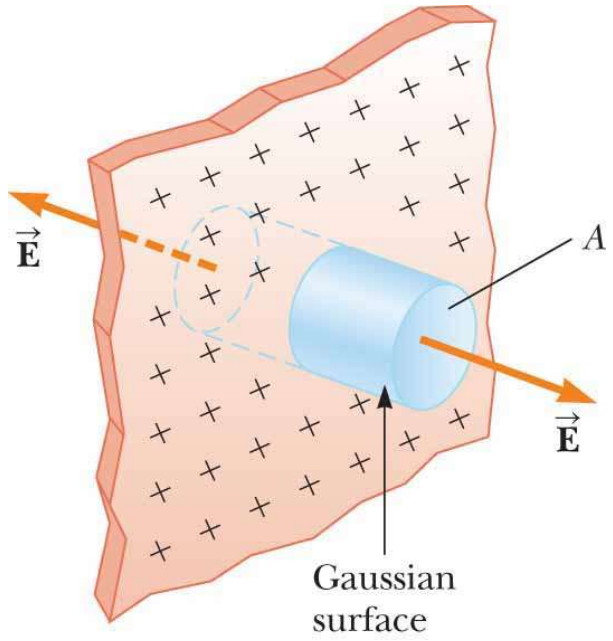
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Example 23.8: A Plane of Charge (1 of 3)

Find the electric field due to an infinite plane of positive charge with uniform surface charge density σ.

$$\Phi_E = 2EA = \frac{q_{\text{in}}}{\varepsilon_0} = \frac{\sigma A}{\varepsilon_0}$$

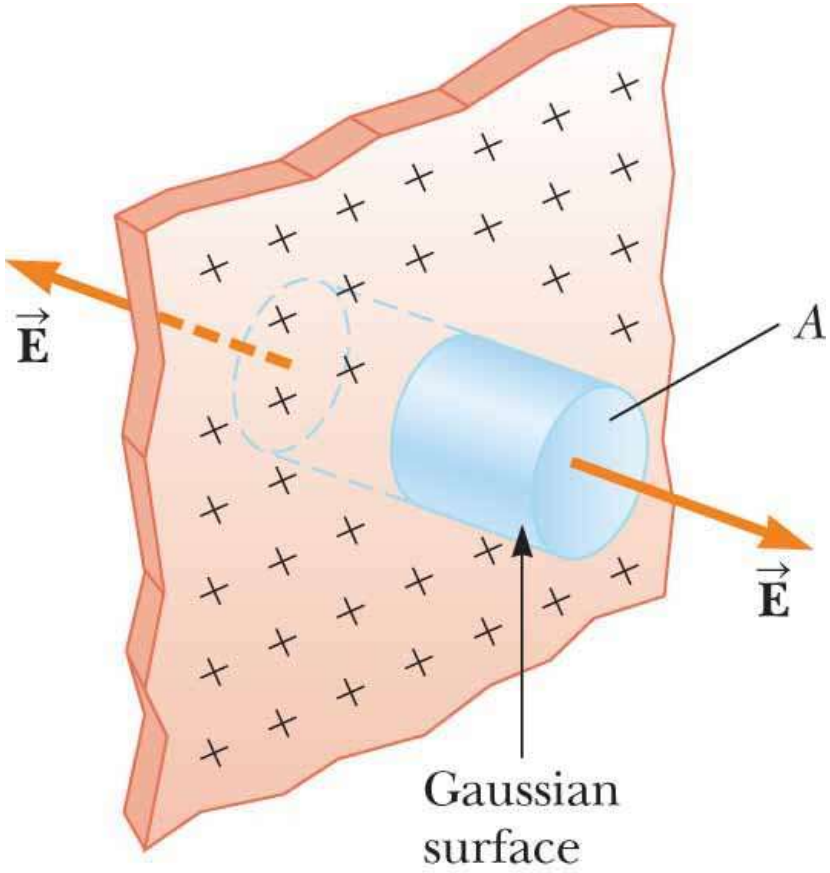
$$E = \frac{\sigma}{2\varepsilon_0}$$

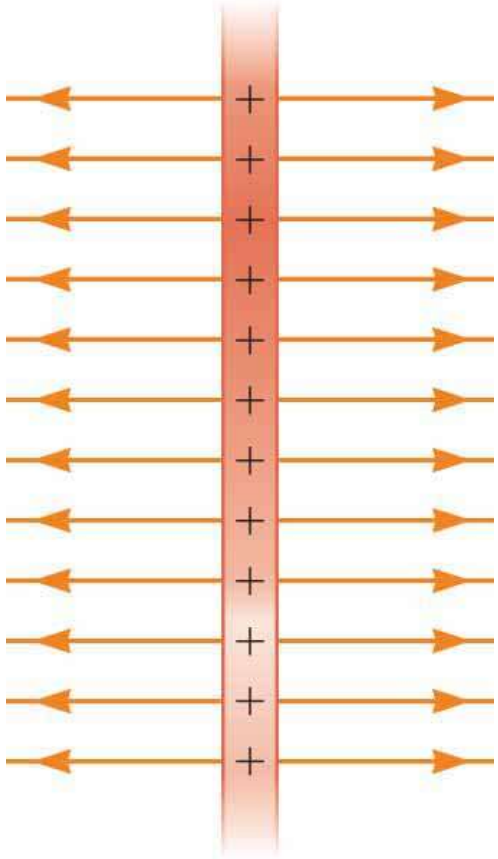


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Example 23.8: A Plane of Charge (2 of 3)



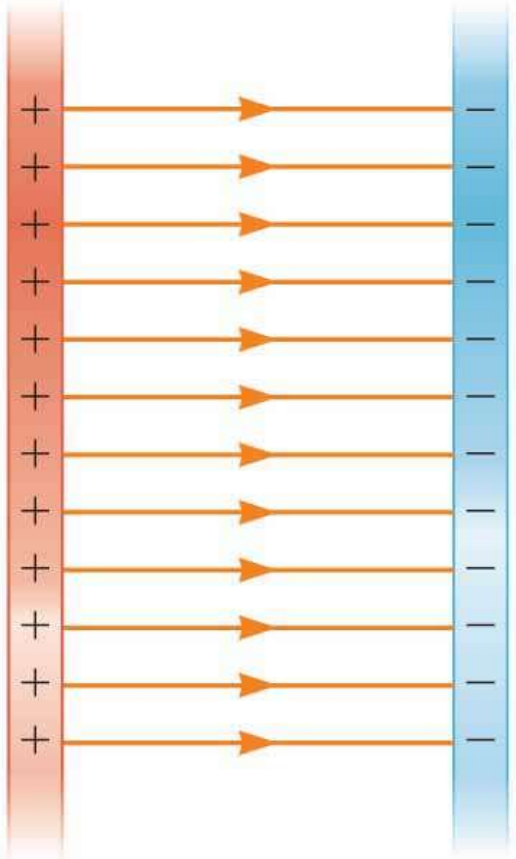


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Example 23.8: A Plane of Charge (3 of 3)

Suppose two infinite planes of charge are parallel to each other, one positively charged and the other negatively charged. The surface charge densities of both planes are of the same magnitude. What does the electric field look like in this situation?



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Conceptual Example 23.9: Don’t Use Gauss’s Law Here!

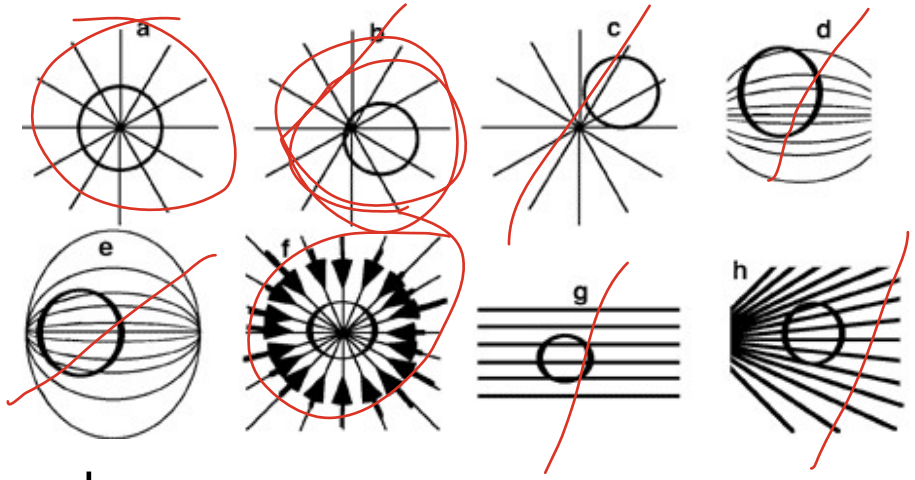
Explain why Gauss’s law cannot be used to calculate the electric field near an electric dipole, a charged disk, or a triangle with a point charge at each corner.

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Assessing to Learn (1 of 4)

The circles in the picture below are Gaussian surfaces. All other lines are electric field lines. For which cases is the flux *non*-zero?



1. a
2. a, b, and f
3. a, b, e, and f
4. a, b, d, e, and h
5. a and b
6. All but g 7. All of them 8. None of the above
7. Cannot be determined

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Assessing to Learn (2 of 4)

We construct a closed Gaussian surface in the shape of a spherical balloon. Assume that a small glass bead with total charge Q is in the vicinity of the balloon. Consider the following statements:

A. If the bead is inside the balloon, the electric flux over the balloon's surface can never be 0.
B. If the bead is outside the balloon, the electric flux over the balloon surface must be 0.
Which of these statements is valid?

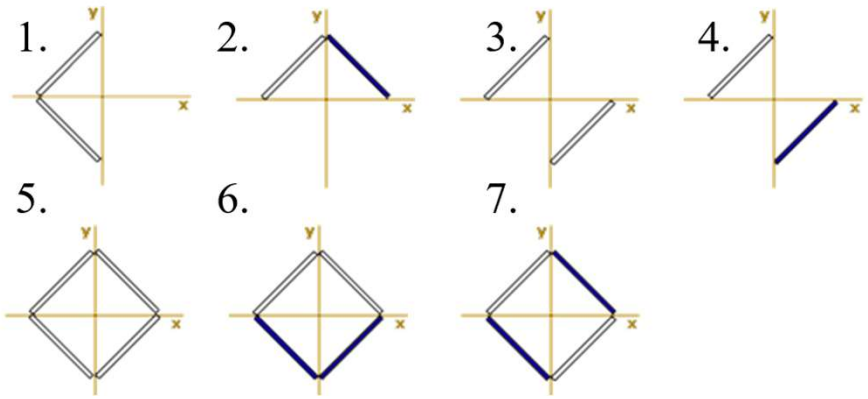
1. Only A is valid.
2. Only B is valid.
3. Both A and B are valid.
4. Neither one is valid.

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Assessing to Learn (3 of 4)

All charged rods have the same length and the same linear charge density (+ or -). Light rods are positively charged, and dark rods are negatively charged. For which arrangement below would the magnitude of the electric field at the origin be largest?



8. Impossible to determine

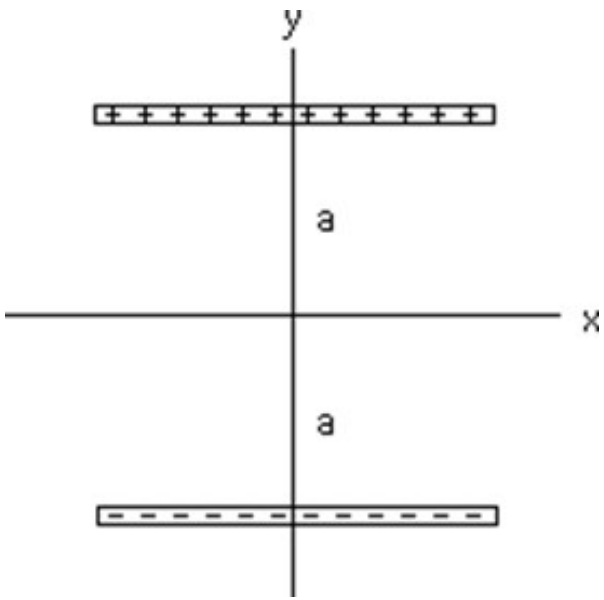
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Assessing to Learn (4 of 4)

Two uniformly charged rods are positioned horizontally as shown. The top rod is positively charged and the bottom rod is negatively charged. The total electric field at the origin:

1. is zero.
2. has both a non-zero *x* component and a non-zero *y* component.
3. points totally in the +*x* direction.
4. points totally in the –*x* direction.
5. points totally in the +*y* direction
6. points totally in the –*y* direction.
7. points in a direction impossible to determine without doing a lot of math.



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